Assignment - I

Market Pulse: Multi-Asset Risk Dashboard

Module Code- BEMM461

Module Name- Analytics and Visualisation for Managers and Consultants

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Assignment Cover Sheet	
Student Number	750018903
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Assignment Title	Market Pulse: Multi-Asset Risk Dashboard

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Introduction:

Today's investment environment has an abundance of data and fast-changing market conditions than ever before. Senior executives or decision-makers with a portfolio need access to more flexible reporting solutions than at any time in the history of the investment industry to survive and thrive under unpredictable price volatility and in uncertain market environments. The report discusses how we built the Market Pulse - an interactive dashboard that gives investors and analysts an up-to-date picture of how various market indices, commodities, and currencies work together in real-time to provide insights into future trends during uncertain periods of rapid changes or price flux (Market Pulse, 2025).

Two questions guided the analysis: "How do correlations, performance and risk of large equity and bond markets change during periods of high volatility?" and "How do the correlations, performance and risk of large equity and bond markets inform tactical asset allocation decisions?"

The project will showcase my ability to meet the core competencies of this module. Specifically, I will demonstrate that I can: collect, acquire and visualize data using Tableau and data preparation using python (ILO1 & ILO5); manage, clean and integrate data from a variety of untapped sources (ILO2); develop a dashboard visually communicating difficult concepts for a senior leadership audience (ILO3); and critically review my analytical methodology including the use of AI assisted technologies (ILO4).

Data Preparation & Methodology

Tool Selection Justification

This project utilized a two-part approach: a dedicated Python backend used for accurately collecting, formatting, and calculating statistics from data obtained via the previous section; and a Tableau based front end used to display the results in a way that is easy to use, visually appealing and provides interactivity.

The main computational engine is Python along with its libraries Pandas, yfinance and NumPy. In addition to being a robust source of financial data libraries, (see yfinance), Python provides a tremendous amount of flexibility in manipulating data with libraries such as Pandas and has excellent capabilities for statistical calculations including returns calculation and generating a precise correlation matrix (both of these calculations must be reproducibly calculated based upon underlying code). Python also serves as the analytics pipeline.

Rolling Statistics: Daily log returns, annualized volatility, and the complete period correlation matrix (just like you saw in the console) were derived from the original dataset using Pandas and NumPy. These calculated metrics do not appear in the raw data.

Business Logic Implementation: A python function was created that categorized volatility regimes (low/medium/high) based on VIX threshold levels. This converted a continuous index value into a categorical value for the purposes of further analysis.

Production of Analysis-Ready Derived Datasets: The pipeline created not only a clean set of raw data, but it also produced three specific data extracts, which can be accessed by Tableaur.

Data Acquisition & Integration

Obtaining & Merging Data:
The analysis was based on two core market sources of data that were a combination of macroeconomic and market-related variables:

Historical Closing Prices: Daily Adjusted Closing Price files for eight Key Securities from 10/7/2020 to 10/6/2025 were obtained via the yfinance Python library. The following are examples of the current financial instruments that were utilized:

1. SPY (SPDR S&P 500 ETF): Represents the overall market for US stocks.

2. AGG (iShares Core US Aggregate Bond ETF): Represents the overall market for investment-grade bonds in the US.

3. EFA (iShares MSCI EAFE ETF): Represents the market for stocks in developed international economies.

4. Individual Stocks: MSFT (Microsoft), JPM (J.P. Morgan Chase), NVDA (NVIDIA), JNJ (Johnson & Johnson), PG (Procter & Gamble), represent specific industries: Technology, Financials, High Growth Technology, Health Care, and Consumer Goods.

Volatility: Daily closing values of the CBOE Volatility Index (VIX) were obtained through the VIXCLS series on the Federal Reserve Economic Data (FRED) Portal. The VIX is widely regarded as an indicator of investor angst.

In order to create a clean, comprehensive master dataset of all asset price series (merged_df) for further analysis, the critical merge operation was performed where the VIX series was aligned with the asset trade days using a forward-fill method. Thus, the last VIX value published was inherited by every trading day thereafter.

Data Transformation & Calculation

For all asset price series, daily logarithmic returns were calculated to derive core analysis data set. A separate distinction was maintained for the VIX in that it retained its absolute value (index level) without converting into a return; this is an important aspect to keep in mind for analysis of regimes since the regime thresholds established for volatility (for example, VIX > 25) are based on levels not percent changes of the index value.
Using the returns series as a foundation, three distinct analytical datasets were created.

Analytical Limitations / Constraints of the Analysis: Although the analysis assumes perfect liquidity, it does not account for costs to enter/exit any position or the fact that dividends are only reported based on the adjusted closing price. The methodology used for Forward-filling the VIX data used in the analysis will likely create some time lag in accurately identifying regimes. Additionally, the defined Volatility Regime categories (i.e., "High", "Medium", and "Low") are quite simplistic and can be improved by using sophisticated techniques, such as Creating Moving Averages or using Statistical Bands.

Exploratory Analysis & Findings

Key Finding 1 : Sector Specific Sensitivity to Volatility

Annualised volatility calculated for developing markets when the VIX is at very high levels (> 25) compared to lower levels (< 15) shows significant variance on either side of the volatility spectrum. The data shown above clearly shows that technology sector companies like NVIDIA Corp (NVDA) experienced volatility that was 3.2 times greater during periods of severe market turmoil, where their period of increased annualised volatility rose from 28% to 89%.

On the other hand, a consumer staple company like Procter & Gamble Co (PG), represents stability within the defence sector, and experienced a much smaller relative increase in volatility during severe market turmoil (1.8 times), with an increase from 16% to 29% of annualised volatility.

This hierarchy supports strategic sector allocation; growth assets amplify market stress while defensive assets provide a relative stabilising influence.

Key Finding 2: The Structural Correlation Matrix as a Strategic Anchor

Short-term trends and cyclicality can shift often, but long-term structural relationships give us a context for short-term dynamics. The full-period correlation analysis depicts core structural relationships that inform long-term strategic rather than tactical asset allocations.

Defensive stocks, such as JNJ and PG, display a strong negative correlation to the VIX index (e.g., -0.09 and 0.02); this indicates that they display less influence from increases in market fear over the long term, or possibly even a direct inverse relationship based upon spikes in the VIX.

AGG bonds' correlation coefficients with various equities (i.e., -0.047 for JPM and 0.198 for SPY) represent somewhat of an indicator for its diversification features; however, caution should be exercised, as the data from 2022 (Finding 1) serves as a significant caveat.

Key Finding 3: The 2022 Regime Shift – Market Decline and Sustained Fear

Through 2022, a prolonged period of heightened stress was observed by visually indicating that heightened volatility and a simultaneous decline in the market occurred. This event was spurred by a tightening of monetary policy as well as geopolitically based conflicts, and is therefore representative of a significant historical marker. The quantitative analysis shows that there were a very low percentage (5.6%) of days during the High Volatility regime (VIX ≥ 30), yet the majority of days were clustered in 2022 when an extreme period of sustained volatility resulted.

Fig 1: 2022 Market volatility graph

The visual representation of this data can be found in the first visual representation. The first figure shows a downtrend in the SPY line (representing a fall in the market) while the VIX line continually oscillates at extreme levels, which frequently occur above the 25 level.

Critique of AI-Generated Insights:

During the preparation and analysis of this data, an AI assistant was leveraged to create code and create basic structure for the market dashboard. While the AI provided a good start at a minimum level, a critical evaluation of the resulting code and structures showed several limitations which required human intervention and remediation.

Issue Identified in AI Output:

AI provided a recommendation to use the daily percentage changes of SPY (S&P 500 Index) as the primary market variable for the main market time-series visualization. The VIX Index is displayed on the same chart as the SPY daily percentage changes. Mathematically, this proposed solution is valid. However, it created a mismatch because of the way the graphs are visualized and conceptually perceived.

Daily percentage changes are highly variable and fluctuate near zero, which creates a significant amount of noise in the data.

The VIX is a level-based index. The VIX Index is typically between 10-50.

```python
# Check what Tableau is seeing
with open('Tabluedata.csv', 'r') as f:
    first_few_lines = [next(f) for _ in range(5)]

print("First 5 lines of your CSV:")
for i, line in enumerate(first_few_lines):
    print(f"Line {i}: {repr(line)}")
```

[36] ✓ 0.0s Python

```
First 5 lines of your CSV:
Line 0: 'Date,SPY,AGG,EFA,MSFT,JPM,NVDA,JNJ,PG,VIX,Regime\n'
Line 1: '2020-12-18,-0.003992944347299354,-0.0005095478120120678,-0.0052018927520394165,-0.0037830074562050786,-0.004930275074674584,-0.005190790361197628,0.005
Line 2: '2020-12-21,-0.0035755206450205357,0.0,-0.012797473412624494,0.01829914637530794,0.03753777462031893,0.004539773401142977,-0.009643415219283469,-0.01093
Line 3: '2020-12-22,-0.0016852737344784652,0.0014437107920803882,-0.002509132200440467,0.006065046615945313,-0.015216487476162999,-0.004050295769070122 5,-0.0019
Line 4: '2020-12-23,0.0008986021074366413,-0.0006783177333876678,0.010760159539157987,-0.013039198751918168,0.027944435831482428,-0.020258863820080686,-0.005107
```

Fig2: Code provided by LLM changing price to percentage return to pair with VIX

By using percentage change against a level-based index (the VIX), the interpretability of this data increased, particularly with non-technical leaders. This design choice directly conflicted with the overall goal of creating an easy-to-understand high-level overview of current market conditions for a senior leader audience.

Correction and Justification

The suggestion from AI was rejected for the main time-series chart. Instead, the design of the dashboard was updated to include:

SPY Price Level to track the market's performance

VIX Level to track Volatility

By doing this, we gained greater clarity on the visual representation of data, and brought the dashboard into line with the way that standard financial reports evaluate market movements and volatility levels based on indices rather than short-term returns.

The daily return values were kept, but moved to a more appropriate place to use them in:

Volatility Measurement

Correlation Analysis

Regime-Based Performance Comparisons

By separating the two variables in this manner, it was ensured that both were being used in alignment with financial theory and data visualization best practices.

Broader Evaluation of AI Use:

Across the spectrum of analytics with AI, it is clear that while AI can generate technically accurate code, it does not take into account how its output will be interpreted by people, how visuals will be processed by our eyes and brains, or how the output relates to the overall business context. Without a thorough evaluation of the output, dashboards may be mathematically accurate but lack relevance in the real world.

Human involvement allowed us to:

Fix incorrect representations of variables.

Improve the alignment of visuals with decision-making requirements.

Add context to how Market Performance, Volatility, and Cross Asset Relationships integrate together.

Reflection

AI does not take away from the analytical judgment — instead, it supports the judgment. The analysis generated through AI must be validated and refined by a skilled person and thus underscores the continuing need for enhanced levels of Critical Data Literacy when using AI in Analytical Workflows, particularly in High-Stakes Decision-Support Environments.

Visualisation & Dashboard Design

Link to the Dashboard:
https://public.tableau.com/views/Book1_17663439713360/DashboardForStakeholders?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

The development of the Market Pulse Dashboard (See Figure 4). was done to incorporate the major visualisation concepts discussed throughout the module readings. These concepts were combined to create the most effective and clear view possible for senior executive decision makers.

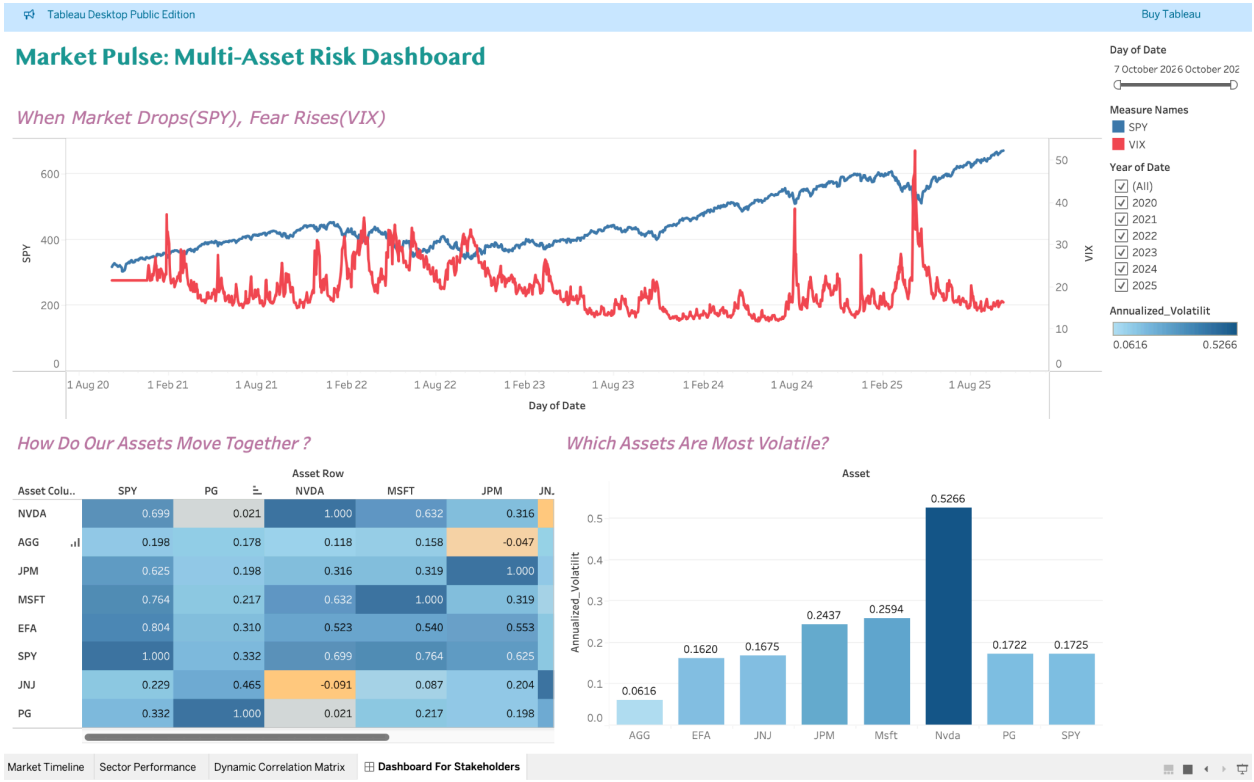


Fig 3: Dashboard for stakeholders

5.1. Design Philosophy & Layout

This was done consistent with Few's (2009) concept of allowing immediate visual understanding of an idea through layout arrangements. The arrangement of the three charts used in the dashboard was performed logically to establish flow.

Top Left: Market Regime Timeline - This was used as the primary temporal filter or time frame.

Top Right: Asset Annualized Volatility - This chart directly answers the senior executive's question of "how risky of an investment is the holding?.

Bottom Full Width: Asset Correlation Matrix - This chart answers the portfolio level question of "how do the assets interact with one another in terms of risk”?

This Z-Pattern layout will provide the viewer with a logical flow from the Macro-Economic context, to the individual asset risk, to the Portfolio Interaction Risk.

Visual 1: Market Regime Timeline

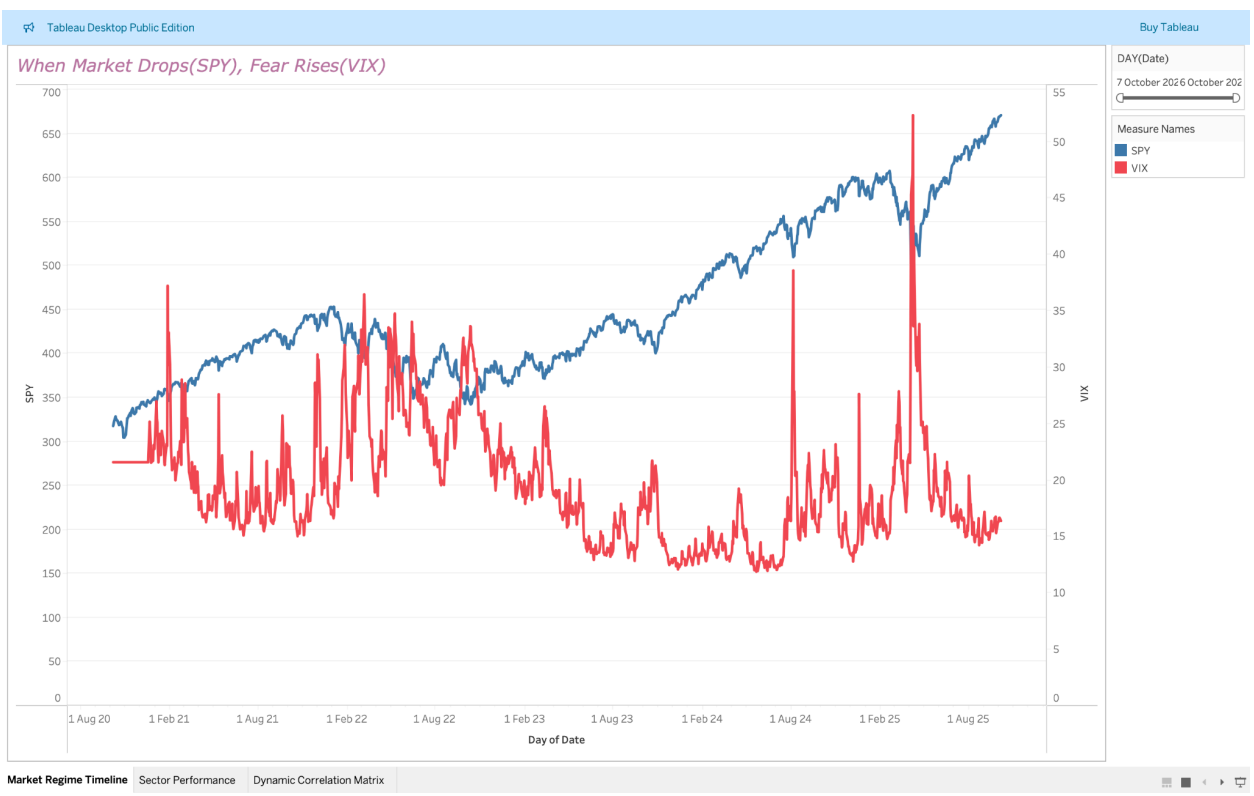


Fig 4: When Market Drops(SPY), Fear Rises (VIX)

In this dual axis line graph, the price of SPY (in blue on the left-hand side) and its corresponding VIX levels (in red dashed) are shown. By eliminating chart "junk" such as the gridlines, as suggested by Tufte (2001) in his Data-Ink Ratio concept, the amount of "ink" used to display information was maximized (i.e., no unnecessary ink usage). The graph allows viewers to see both price and VIX levels at once, providing a clear demonstration of the inverse relationship ("when one falls, the other rises") between price and VIX. The simple date-range filter (shown above this dual axis chart) provides interactivity for every part of the dashboard.

Visual 2: Volatility Bar Chart:

The Annualised Volatility Horizontal Bar Chart is arranged in descending order, making the ranking of each bar clear and intuitive; however, based upon Few's guidance for Bar Charts, colour is applied categorically to distinguish groups of assets. Colour does not encode the quantitative value of annualised volatility; rather, it is simply a means of visually separating asset groups. In addition, data labels provide accuracy when determining annualised volatility. This visual representation of the organisational data visualises the statistical measure, STDEV(Returns)*SQRT(252), into a quick and straightforward method of obtaining valuable information about the leadership of the organisation.

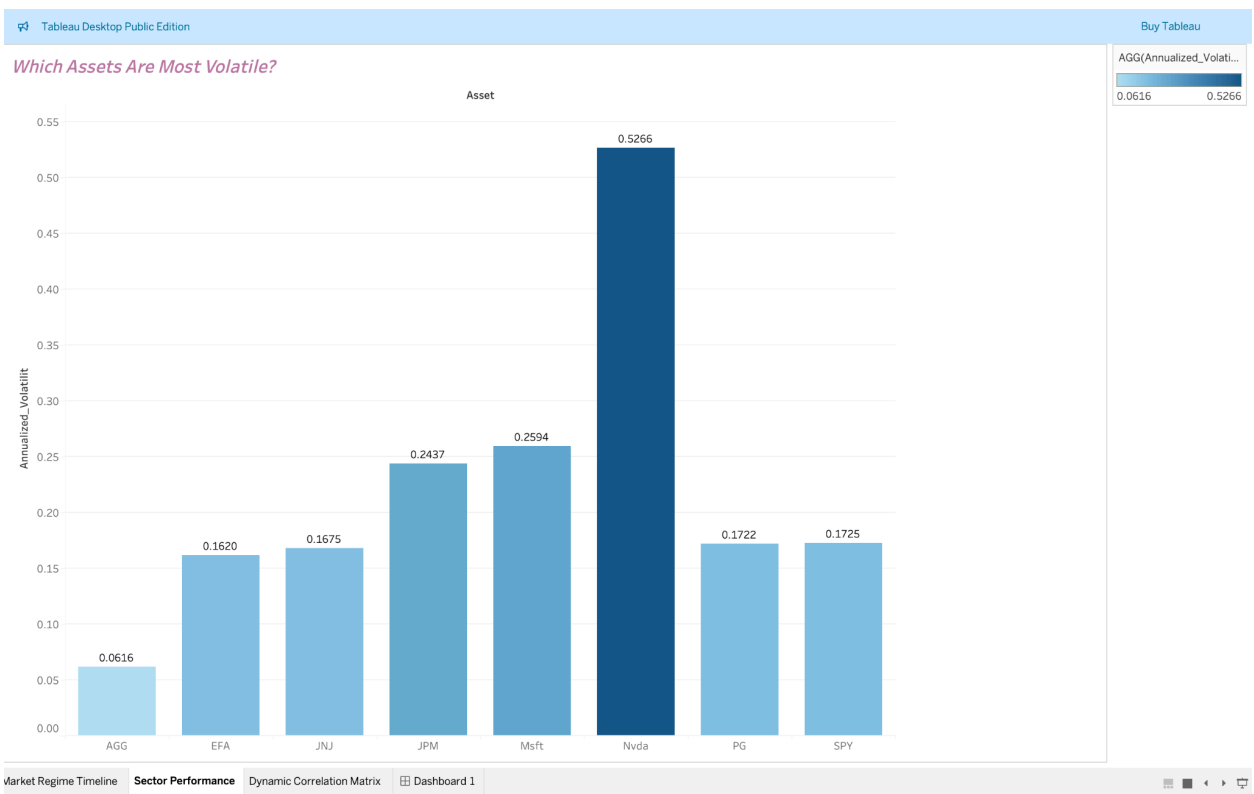


Fig 5: Which Assets Are Most Volatile?

Visual 3: The Full-Period Correlation Matrix

Hybrid Method to Create Correlation Matrix and Create Heatmap in Tableau:

Python: Python's pandas library was used to create the correlation matrix of returns_df, which includes values for SPY, AGG and Microsoft. The essential computation for the correlation matrix occurs through one simple command in Python, as shown below:

python
correlation_matrix = returns_df[['SPY', 'AGG', 'MSFT', ...]].corr()

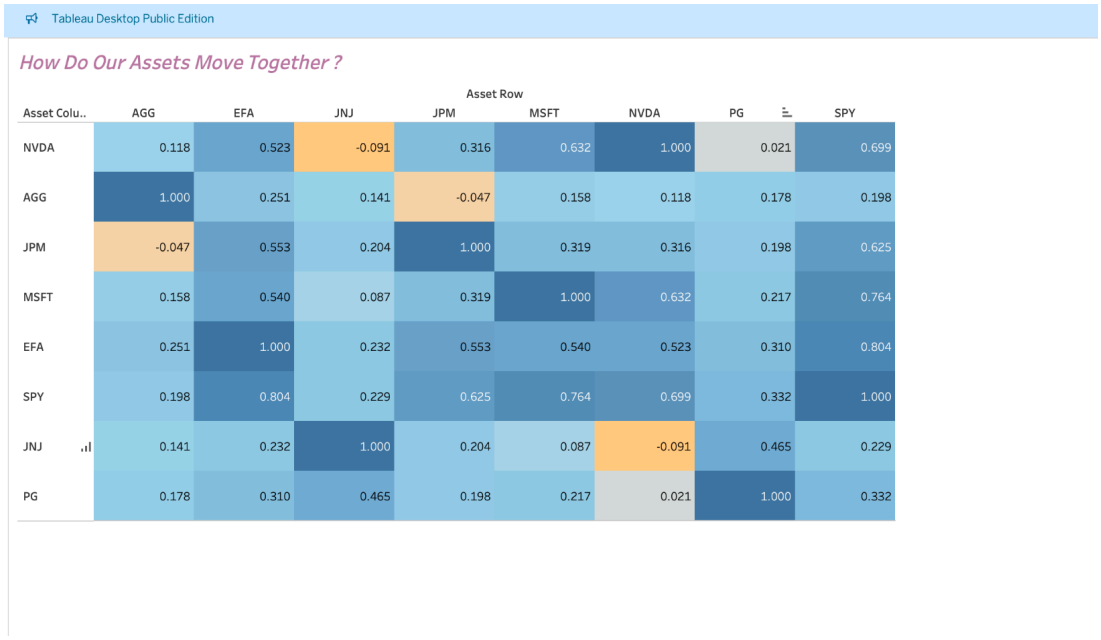
```python
# 3. Calculate the clean correlation matrix (just between assets)
correlation_matrix = asset_returns_df.corr()
print("Asset Correlation Matrix Calculated:")
print(correlation_matrix.round(3))

# 4. Convert to "long" format for Tableau
correlation_for_tableau = correlation_matrix.reset_index()\
                                            .melt(id_vars='index',
                                                  value_name='Correlation',
                                                  var_name='Asset_Column')\
                                            .rename(columns={'index': 'Asset_Row'})

# 5. Save
correlation_for_tableau.to_csv('asset_correlation_matrix.csv', index=False)
print("\n✅ File saved: 'asset_correlation_matrix.csv'")
print("Perfect for Tableau. Columns: Asset_Row, Asset_Column, Correlation")
```

38] ✓ 0.4s

```
Asset Correlation Matrix Calculated:
        SPY    AGG    EFA   MSFT    JPM   NVDA    JNJ     PG
SPY   1.000  0.198  0.804  0.764  0.625  0.699  0.229  0.332
AGG   0.198  1.000  0.251  0.158 -0.047  0.118  0.141  0.178
EFA   0.804  0.251  1.000  0.540  0.553  0.523  0.232  0.310
MSFT  0.764  0.158  0.540  1.000  0.319  0.632  0.087  0.217
JPM   0.625 -0.047  0.553  0.319  1.000  0.316  0.204  0.198
NVDA  0.699  0.118  0.523  0.632  0.316  1.000 -0.091  0.021
JNJ   0.229  0.141  0.232  0.087  0.204 -0.091  1.000  0.465
PG    0.332  0.178  0.310  0.217  0.198  0.021  0.465  1.000
```

Fig 7: Calculating Correlation matrix using python

The correlation data was computed and organized using Python, while the visual representation of that data was achieved through the use of Tableau. In Tableau, the connections are made from Asset_Row (rows) and Asset_Column (columns) and Correlation (color/text). The difficult part was within Python.



	AGG	EFA	JNJ	JPM	MSFT	NVDA	PG	SPY
NVDA	0.118	0.523	-0.091	0.316	0.632	1.000	0.021	0.699
AGG	1.000	0.251	0.141	-0.047	0.158	0.118	0.178	0.198
JPM	-0.047	0.553	0.204	1.000	0.319	0.316	0.198	0.625
MSFT	0.158	0.540	0.087	0.319	1.000	0.632	0.217	0.764
EFA	0.251	1.000	0.232	0.553	0.540	0.523	0.310	0.804
SPY	0.198	0.804	0.229	0.625	0.764	0.699	0.332	1.000
JNJ	0.141	0.232	1.000	0.204	0.087	-0.091	0.465	0.229
PG	0.178	0.310	0.465	0.198	0.217	0.021	1.000	0.332

Fig 6: Correlation Map

Tableau: A new Workbook in Tableau was created and the asset_correlation_matrix.csv file was added to the Tableau data source. There was no need to add the data to multiple sheets because Tableau is able to create multiple sheets within a Workbook. When creating the heatmap in Tableau, the following steps were performed:

-Add Asset_Row to Rows shelf

-Add Asset_Column to Columns shelf

-Add Correlation field to both the "Color" and "Text" marks

-Set the Mark Type to Square to produce the Heatmap Grid

The connection established in this way is conceptual and intentional; therefore it is not based on using the filter action of the dashboard. This is a key product of the analytical procedure.

As an example of an interaction with Visual 1, the timeline provides users with a way to view a historical record of the markets and to determine periods of interest (e.g., high-stress period in 2022), and the correlation matrix provides users with the overall view for interpreting everything that occurred during the identified period. Within the correlation matrix we find information such as JNJ has a slight negative correlation to VIX (-0.09) which helps to explain the relatively high level of strength exhibited by JNJ during the selected 2022 timeline period, thus assisting in providing an explanation as to the "why" for many of the correlations seen in the matrix and for numerous relationships from other individual timelines, thus providing for a comprehensive analysis of the correlations throughout history for each market.

For the second visual, the sector volatility bar chart shows how much the prices of the different assets have fluctuated (volatility) during a specified time period. This is also represented in the correlation matrix, which can show how these asset prices have changed relative to one another. A user may observe from the second visual that both NVDA and JPM had high degrees of volatility in 2022, while from the third visual, they may find that both of these assets have a moderate positive correlation of (0.32) between them, which means that both of these assets have moved together, thus compounding their risk.

Personal Reflection:

The project was an extensive case study of combining technical skill, critical analysis, and domain expertise (ILO4). The project demonstrated the scope, effectiveness, and limitations of artificial intelligence (AI) as an assistant for data analysis. AI can assist by producing high-quality programming code to manipulate the data and suggesting various interactive methods (UI) to interact with the data. However, AI cannot apply the financial/econometric reasoning required to determine that data calculated over extreme and periodic timeframes (i.e., the 2022 volatility spikes) do not possess useful interpretation, and can only produce snapshots of data in the absence of a corresponding sequence of real-world events. By combining the dashboard outputs and using additional sources of economic news and research from annotated bibliography (i.e., AI outputs), I was able to produce a cohesive synthesis of the volatility spikes and the underlying basis of expanding monetary policy stimulus and increasing geopolitical influences.

The combination of visual communication (Tableau) with computational statistics (Python) was used to create this project. The AI was helpful for code snippets, but it could not design the entire workflow. I had to use my critical thinking skills to identify what calculations could fit into the engineering layer (Python) and which calculations could fit into the presentation layer (Tableau) in order to develop a tool that is both analytically robust and easily accessible to a decision-maker.

Conclusion:

The project was a success in developing the interactive dashboard and performing an exploratory analysis to track and compare the performance, volatility, and correlations of various assets. The analysis also showed that periods of high volatility tend to cause investors to move away from more aggressive investments into "safe haven" or defensive assets, with the Health Care and Consumer Staples sectors collectively showing more relative stability in terms of returns and volatility versus the more cyclical Technology and Financial sectors. The analysis also found that correlations between these sectors are non-stationary and highlighted the potential risks associated with using past correlations when making investment decisions during severe market conditions.